PI discussion

Physical diffusion models with fast hidden units

From a local energy model to global digit structure

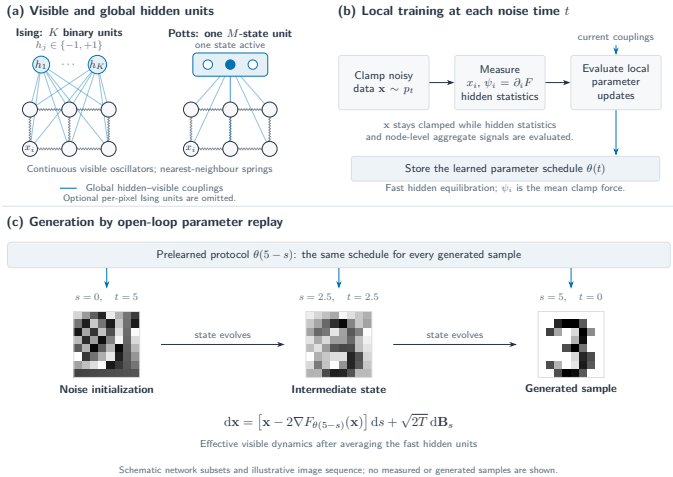
Slow visible oscillators $+$ fast hidden statistics

Local training $\longrightarrow$ open-loop generation

Theory, all-ten-digit MNIST samples, learned schedules and a trajectory movie

11 November 2026

The idea: train local interactions, replay a schedule



Architecture

Continuous pixels with neighbour springs and global hidden coupling.

Training

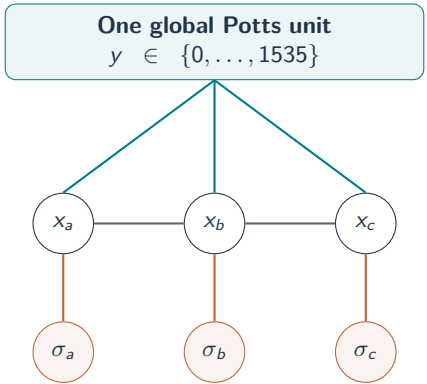
Clamp noisy data; use mean forces and conditional hidden statistics.

Generation

Replay the learned parameters as a function of time.

Overview is schematic. Optional local Ising units are omitted here.

The Potts run we are actually inspecting



Representative pixels; the global Potts unit touches all 784.

- 784 continuous visible oscillators, 28×28 pixels.
- One shared Potts label with 1536 states.
- 784 additional hidden Ising spins, one per pixel.
- Up to eight visible neighbours; 2970 distinct spring edges.
- Quartic on-site potentials; quadratic Potts coupling enabled.

All ten digits. Training uses exact conditional hidden statistics.

Eliminating hidden states gives a free energy

At one fixed diffusion time, let $E(\boldsymbol{x}, y)$ be the joint energy.

$$Z_h(\boldsymbol{x}) = \sum_y e^{-E(\boldsymbol{x},y)/T}, \qquad p(y \mid \boldsymbol{x}) = \frac{e^{-E(\boldsymbol{x},y)/T}}{Z_h(\boldsymbol{x})},$$

$$F(\boldsymbol{x}) = -T \log Z_h(\boldsymbol{x}), \qquad p_\theta(\boldsymbol{x}) \propto e^{-F(\boldsymbol{x})/T}.$$

The identity that connects statistics to mechanics

$$\boxed{-\nabla F(\boldsymbol{x}) = \left\langle -\nabla_{\boldsymbol{x}} E(\boldsymbol{x}, y) \right\rangle_{y|\boldsymbol{x}}}$$

$F(\boldsymbol{x}) = \langle E \rangle_{y|\boldsymbol{x}} - TS(y \mid \boldsymbol{x})$ includes the hidden-state entropy.

The static marginal is exact. Using this mean force dynamically requires fast hidden equilibration.

Fast hopping can produce a smooth average force

$$E_m(\boldsymbol{x}) = E^{\mathrm{v}}(\boldsymbol{x}) - a_m(\boldsymbol{x}), \qquad \boldsymbol{f}_m = -\nabla E_m.$$

$$\begin{aligned} -\nabla F &= -\nabla E^{\mathrm{v}} + T\nabla \log \sum_m e^{a_m/T} \\ &= -\nabla E^{\mathrm{v}} + \sum_m p_m \nabla a_m \\ &= \sum_m p_m \boldsymbol{f}_m. \end{aligned}$$

For example

Spend 70% of the time in state 1 and 30% in state 2:

$$\overline{\boldsymbol{f}} = 0.7\boldsymbol{f}_1 + 0.3\boldsymbol{f}_2.$$

Hidden mixing must be fast compared with visible motion *and* changes in the parameter schedule.

The simulation evaluates the average directly. It contains no discrete hidden jumps or switching rates.

Potts: one label selects a state-dependent landscape

$$E_{\mathrm{eff}}(\boldsymbol{x}, y = m) = G(\boldsymbol{x}) - a_m(\boldsymbol{x}),$$
$$a_m(\boldsymbol{x}) = b_m + \boldsymbol{W}_m \cdot \boldsymbol{x} + \boldsymbol{Q}_m \cdot \boldsymbol{x}^2,$$
$$p_m(\boldsymbol{x}) = \frac{e^{a_m / T}}{\sum_k e^{a_k / T}},$$
$$\boldsymbol{f}_{\mathrm{Potts}} = \sum_m p_m(\boldsymbol{x}) \big(\boldsymbol{W}_m + 2 \boldsymbol{Q}_m \circ \boldsymbol{x} \big).$$

$G = E^{\mathrm{v}} + F_{\mathrm{local}}$ includes the local Ising contribution when present.

W_m**: linear tilt**
Changes the preferred visible configuration.

Q_m**: stiffness change**
It is not the covariance itself.

b_m**: energy offset**
Changes relative occupation, not spatial position.

A mixture of M conditional visible distributions. They are Gaussian only for a quadratic common energy.

Ising: binary hidden units can act in combination

$$h_j \in \{-1, +1\}, \quad j = 1, \dots, K,$$

$$E_{\text{eff}}(\boldsymbol{x}, \boldsymbol{h}) = G(\boldsymbol{x}) - \sum_j h_j a_j(\boldsymbol{x}),$$

$$F(\boldsymbol{x}) = G(\boldsymbol{x}) - T \sum_j \log\left[2 \cosh \frac{a_j(\boldsymbol{x})}{T}\right],$$

$$\langle h_j \mid \boldsymbol{x} \rangle = \tanh \frac{a_j(\boldsymbol{x})}{T},$$

$$\boldsymbol{f}_{\text{Ising}} = \sum_j \tanh(a_j / T)\, \nabla a_j.$$

Global Ising architecture

Each spin couples to all pixels. K binary spins give 2^K joint configurations.

Additional local spins

One per pixel: $a_i^{\text{loc}} = c_i + w_i x_i$.

$$f_i^{\text{loc}} = w_i \tanh \frac{c_i + w_i x_i}{T}.$$

Training uses clamped noisy data and hidden statistics

At each noise time t, fit F_θ to the score of p_t:

$$\mathcal{J}_T(\theta) = \mathbb{E}_{\boldsymbol{x}\sim p_t}\left[-\frac{\Delta F_\theta}{T} + \frac{\|\nabla F_\theta\|^2}{2T^2}\right].$$

Visible parameters: $E^{\mathrm{v}} = \sum_k \theta_k \phi_k$

$$\partial_{\theta_k}\mathcal{J}_T = \left\langle -\frac{\Delta\phi_k}{T} + \frac{\nabla\phi_k\cdot\boldsymbol{\psi}}{T^2}\right\rangle,$$

where $\boldsymbol{\psi} = \nabla F$ is the mean clamp force.

Hidden parameters

Updates combine pixel signals, conditional occupations/moments, current couplings and state-level aggregates.

No free-running negative phase is required.

The all-ten-digit results here use exact conditional statistics. Finite-read hardware training is a separate test.

Generation replays the learned parameters open loop

$$\text{Forward noise:} \quad d\boldsymbol{x} = -\boldsymbol{x}\,dt + \sqrt{2T}\,d\boldsymbol{B}_t,$$

$$\text{Generation:} \quad s = 5 - t,$$

$$\boxed{d\boldsymbol{x} = \left[\boldsymbol{x} - 2\nabla F_{\theta(5-s)}(\boldsymbol{x})\right] ds + \sqrt{2T}\,d\boldsymbol{B}_s.}$$

- One learned schedule $\theta(t)$, linearly interpolated between 100 time slices.
- The same schedule for every generated image; no training during generation.
- Hidden probabilities respond to the evolving $\boldsymbol{x}$ through the couplings.

The effective potential for generation is $2F - \|\boldsymbol{x}\|^2/2$; physical realization must reproduce that force.

All ten digits: saved samples from both architectures

One Potts unit, $M = 1536$

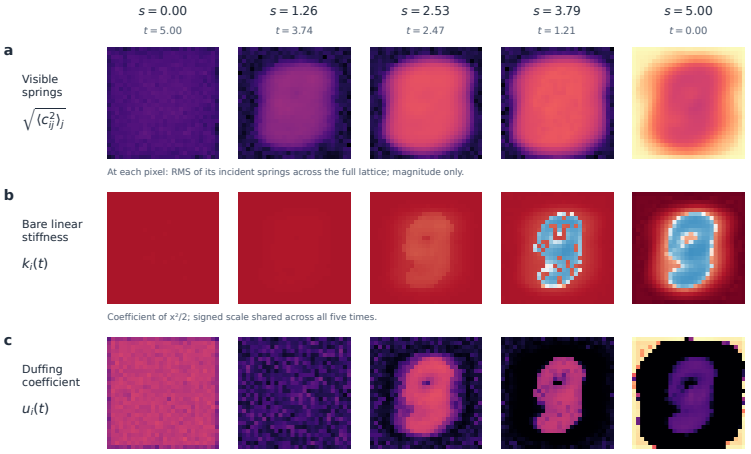


Global Ising, $K = 784$



First 100 images of each saved base-2000 sample set, unfiltered and in saved order. Both include local Ising units and eight-neighbour visible springs. These are examples, not a controlled ranking.

Visible parameters develop strong spatial structure



Top: springs
RMS of incident c_{ij}.

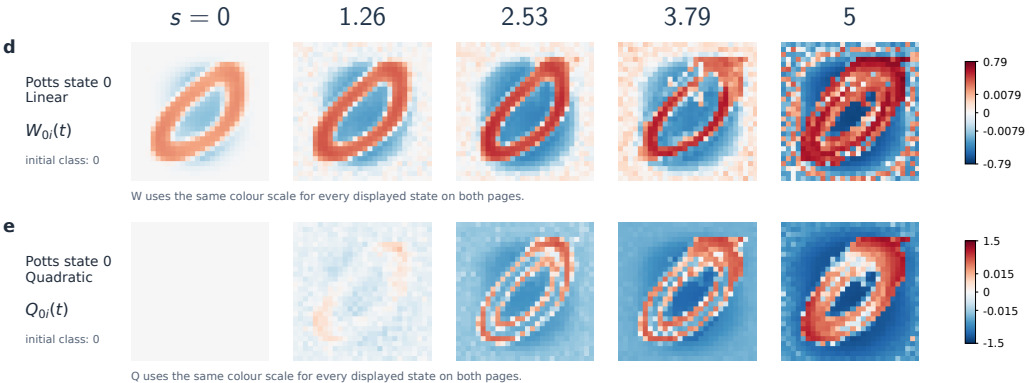
Middle: k_i
Bare coefficient of $x_i^2/2$.

Bottom: u_i
Duffing coefficient of $x_i^4/4$.

Log magnitudes; signed k_i uses symmetric log.

Potts1536 diagnostic run. Columns follow generation time $s = 5 - t$. Colour scales are shared across time within each row.

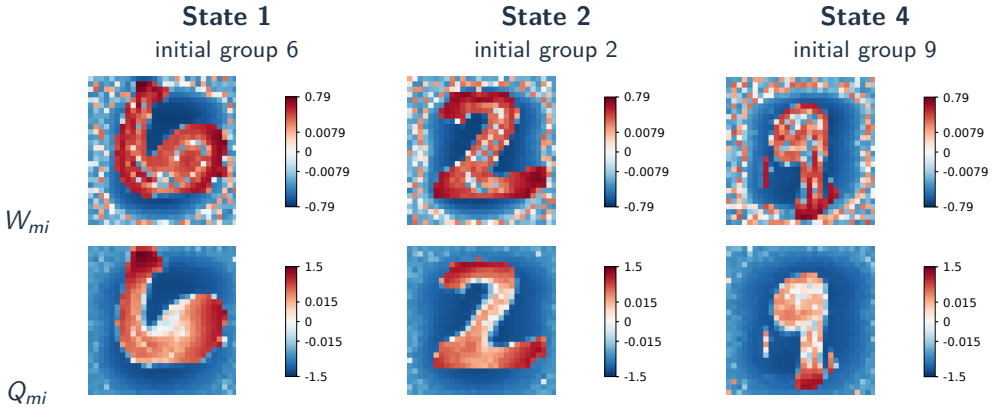
One Potts state's coupling pattern evolves with time



W_{0i} supplies a linear field; Q_{0i} changes local stiffness. The pattern resembles a prototype.

State 0, whose initial cluster majority label is 0. Raw coefficients; common symmetric-log scales across states and times.

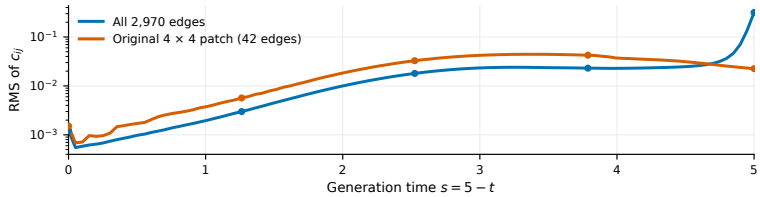
Other Potts states carry different spatial patterns



Final-time couplings, $s = 5$. Initial cluster labels are approximate associations, not measured final class purity.

Crops of the existing parameter figure. Each coefficient family uses the same scale for all states. Full schedules are in backup.

The central spring patch is not the whole lattice



Peak spring magnitude
Full lattice: $s = 5.00$
RMS = 0.317

Central patch: $s = 3.43$
RMS = 0.0443

Central 4×4 patch

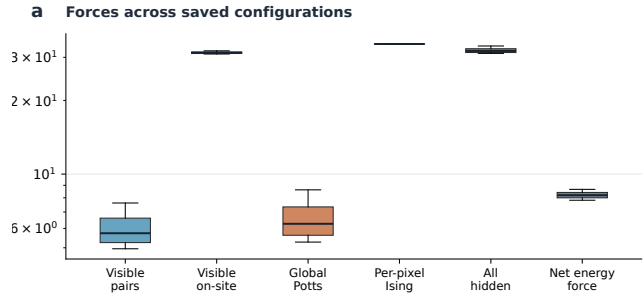
42 edges; peak at $s \simeq 3.43$.
The spring RMS decreases near the data end.

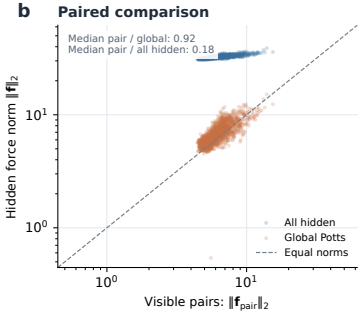
Full visible lattice

2970 edges; maximum at $s = 5$.
Strong boundary structure dominates the late rise.

Springs suppress differences between pixels. They do not, by themselves, fix the absolute background amplitude.

Visible-pair and global-Potts forces are comparable

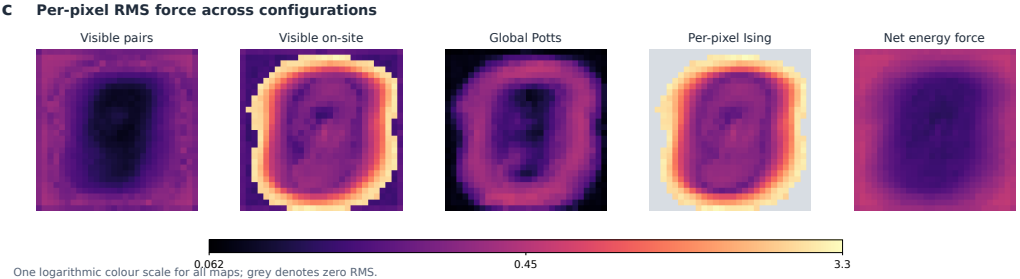




Median norms: visible pairs 5.73, global Potts 6.27; on-site 31.30, local Ising 34.01.

All 2000 saved final configurations, $s = 5$; no quality filtering. Norms are calculated across pixels separately for each image.

Large visible and hidden forces mostly oppose each other



Alignment

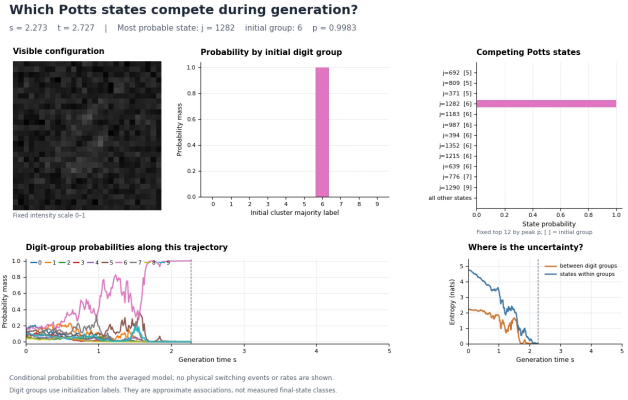
Median $\cos\angle(\boldsymbol{f}_{\mathrm{vis}}, \boldsymbol{f}_{\mathrm{hid}}) = -0.968.$

Cancellation

Median $\dfrac{\|\boldsymbol{f}_{\mathrm{vis}} + \boldsymbol{f}_{\mathrm{hid}}\|}{\|\boldsymbol{f}_{\mathrm{vis}}\| + \|\boldsymbol{f}_{\mathrm{hid}}\|} = 0.126.$

Spatial RMS across the same 2000 final configurations. Visible = pairs + on-site; hidden = global + local. One logarithmic colour scale.

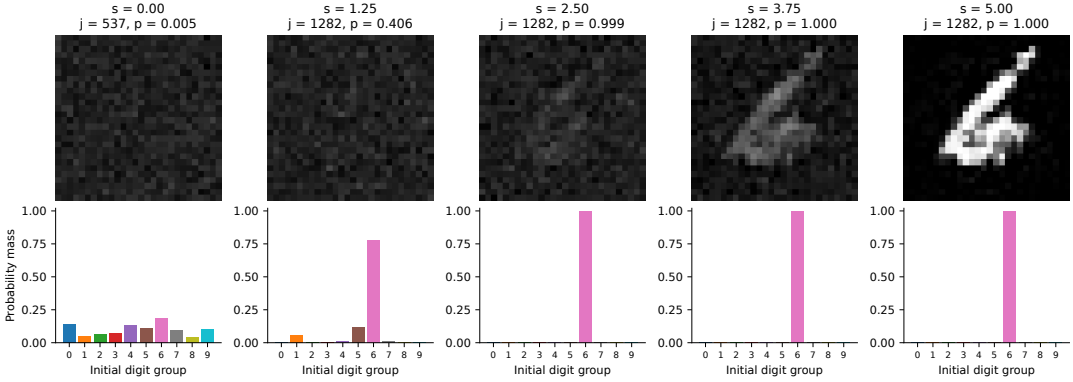
Movie: one visible trajectory and its Potts probabilities



Open attached MP4 Open companion MP4

20 seconds; 397 saved frames. One actual reverse-SDE trajectory. Curves show conditional probabilities, not simulated hidden jumps.

State confidence concentrates before the digit is clear



In this trajectory: group confidence exceeds 90% from $s \simeq 1.65$; individual-state confidence from $s \simeq 1.94$.

Thresholds remain exceeded at all subsequent saved frames. Final state 1282 has initial group 6. One illustrative trajectory, not a population study.

What is established, and what should come next?

Established numerically

- Trainable visible and hidden energy parameters.
- All-ten-digit generation in the conditional-mean model.
- Structured coupling schedules and interpretable force contributions.

Follow-up directions

- Explicit hidden dynamics and finite switching rates.
- Finite-read learning and architectural ablations.
- Scaling, encoder/decoder interfaces and hardware constraints.

Proposed paper focus: local training and a working latent extension. Treat physical kinetics and scaling as follow-up studies.

Backup: the Gaussian special case

If the common energy is quadratic,

$$G(\boldsymbol{x}) = \tfrac{1}{2}\boldsymbol{x}^T A\boldsymbol{x} + \ell^T \boldsymbol{x}, \qquad a_m = b_m + \boldsymbol{W}_m^T \boldsymbol{x} + \boldsymbol{Q}_m^T \boldsymbol{x}^2,$$

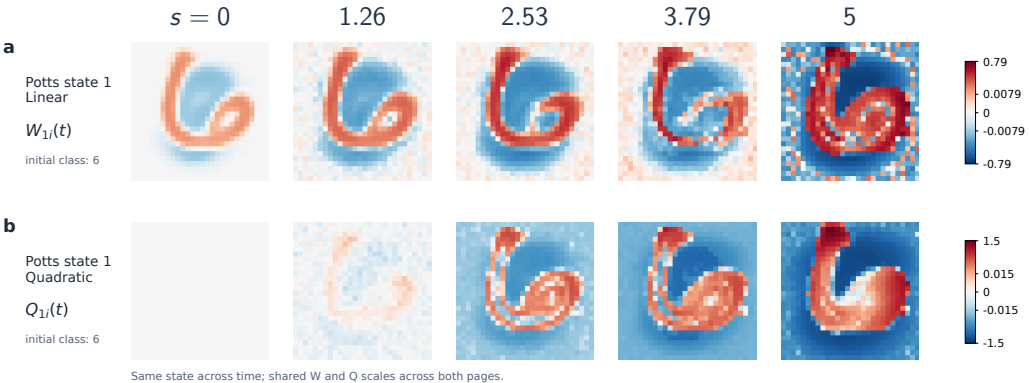
define $H_m = A - 2\,\mathrm{diag}(\boldsymbol{Q}_m)$. For $H_m \succ 0$,

$$\boxed{p(\boldsymbol{x} \mid m) = \mathcal{N}(\boldsymbol{\mu}_m, \Sigma_m), \quad \boldsymbol{\mu}_m = H_m^{-1}(\boldsymbol{W}_m - \ell), \quad \Sigma_m = TH_m^{-1}.}$$

- Q_m changes precision/stiffness; visible springs also affect covariance.
- W_m sets the linear tilt; b_m cancels from normalized $p(\boldsymbol{x} \mid m)$.
- Mixture weights also depend on each component's normalization.

The actual quartic-plus-local-Ising model is more general than this Gaussian special case.

Backup: state 1, initially associated with digit 6

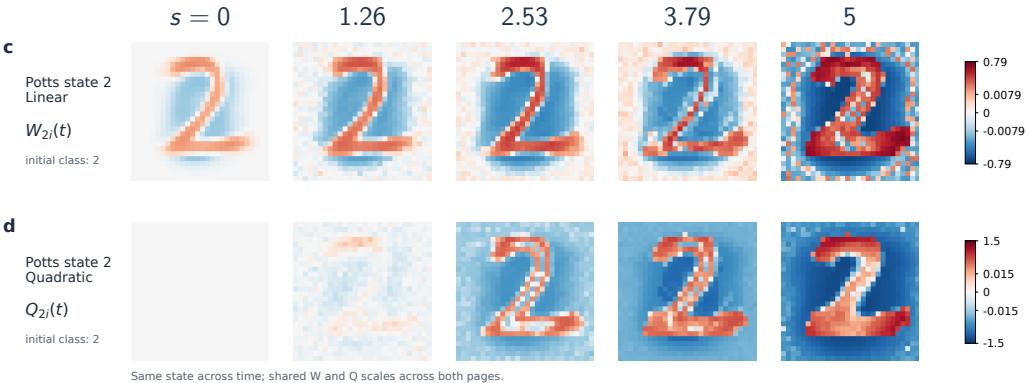


Same state across time; shared W and Q scales across both pages.

State 1. Shared W and Q colour scales across all shown states and times.

Saved coefficients; $s = 5 - t$. Initial digit associations describe the initialization clusters.

Backup: state 2, initially associated with digit 2

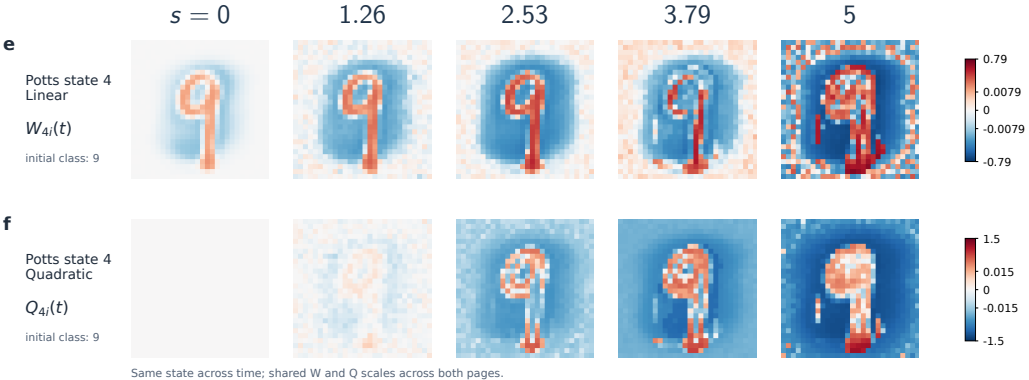


Same state across time; shared W and Q scales across both pages.

State 2. Shared W and Q colour scales across all shown states and times.

Saved coefficients; $s = 5 - t$. Initial digit associations describe the initialization clusters.

Backup: state 4, initially associated with digit 9

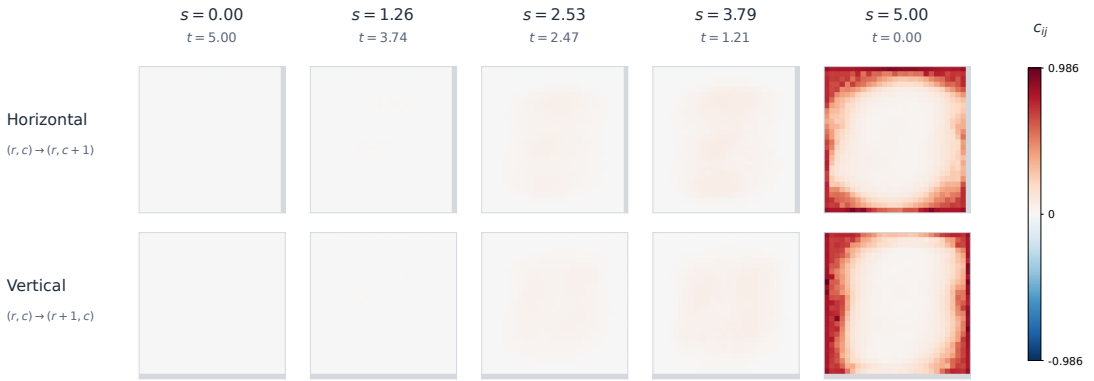


Same state across time; shared W and Q scales across both pages.

State 4. Shared W and Q colour scales across all shown states and times.

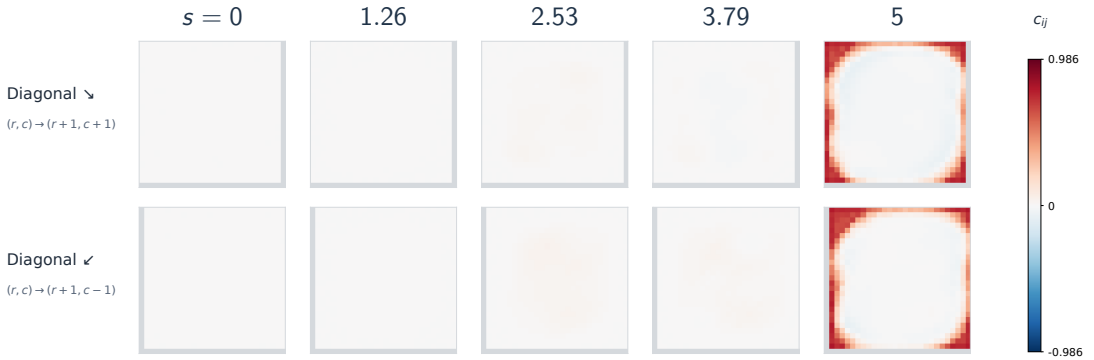
Saved coefficients; $s = 5 - t$. Initial digit associations describe the initialization clusters.

Backup: signed horizontal and vertical spring maps



Each map location represents one edge. A shared linear colour scale makes weak early couplings look nearly white.

Backup: signed diagonal spring maps



Diagonal edges complete the eight-neighbour lattice; boundary pixels have fewer neighbours.